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Section ÷ B

Subject ÷ Wireless communication

Assignment - 2

1 If 0 dBm is equal to 1 mW (10^{-3} W) over a 50 Ω load; express 10 W in units of dBm.

→ Formula $= P(\text{dBm}) = 10 \log_{10} \left(\frac{P(\text{mW})}{1 \text{ mW}} \right)$

Given :

$$P = 10 \text{ W}$$

Converting watts into milliwatts :

$$10 \text{ W} = 10 \times 1000 = 10,000 \text{ mW}$$

Therefore,

$$P(\text{dBm}) = 10 \log_{10} \left(\frac{10000}{1} \right)$$

$$\therefore P(\text{dBm}) = 10 \times 4 = 40 \text{ dBm} \#$$

2. A total of 24 MHz of bandwidth is allocated to a particular FDD cellular telephone system that uses two 30 kHz simplex channels to provide full duplex voice and control channels.

a) Find the number of channels in each cell for a four-cell reuse system.

Given

Total bandwidth = 24 MHz

One simplex channel = 30 kHz

Two simplex channel = one duplex channel

Reuse factor (N) = 4

Traffic / user = 0.1 Erlang

$$\text{Total no. of duplex channels} = \frac{24,000 \text{ kHz}}{2 \times 30 \text{ kHz}}$$

$$= \frac{24000 \text{ kHz}}{60 \text{ kHz}}$$

$$= 400$$

$\therefore$ Total number of channels is 400.

For a four-cell reuse pattern,

$$\text{channel/cell} = \frac{400}{4}$$

$$= 100$$

b. If each cell is to offer capacity that is 90% of perfect scheduling, find the maximum number of users that can be supported per cell where omnidirectional antennas are used at each base station.

→ Assume, 90% of the available 100 channels are used
i.e. $100 \times 0.90 = 90$.

∴ offered traffic = approx. 90 Erlangs,
∴ Each user generates 0.1 Erlangs,

$$\text{Users} = \frac{90}{0.1}$$

$$= 900 \text{ users/cell}$$

c. What is the blocking probability of the system in (b) when the maximum numbers of users are available in the user pool?

→ For 100 channels and approx. 90 erlangs offered traffic, Erlangs B gives,

$$P_b = \frac{A^N / N!}{\sum_{k=0}^N A^k / k!}$$

for, $N = 100$, $A = 90$

The blocking probability is approximately,

$$P_b \approx 2.70\%$$

$\therefore$ Approximately 2.7% of calls are blocked at this traffic level.

d. If each new cell now uses 120° sectoring instead of omnidirectional for each base station, what is the new total number of users that can be supported per cell for the same blocking probability as in (c)?

→ With 120° sectoring, each cell is divided into three sectors with total channel per cell 100 and approximately,

$$\frac{100}{3} \approx 33 \text{ channels per sector}$$

Using the same blocking probability of approximately 2.70%, Erlangs B gives approximately ~~2.70~~ 25.5 Erlangs per 33-channel sector.

For three sectors,

$$A_{\text{total}} = 25.5 \times 3$$

$$A_{\text{total}} \approx 76.5 \text{ Erlangs}$$

$$\text{Users} = \frac{76.5}{0.1} = 765 \text{ users/cell}$$

e. If each cell covers five square kilometers, then how many subscribers could be supported in an urban market is $50\text{ km} \times 50\text{ km}$ for the case of omnidirectional base station antennas?

$$\rightarrow \text{Area} = (50 \times 50) \text{ km}^2 = 25,000 \text{ km}^2$$

Each cell covers $\div 5 \text{ km}^2$

$$\text{No. of cells} = \frac{25,000}{5} = 500$$

$$\text{For omnidirectional antennas} = 500 \times 900 \\ = 450,000 \text{ subscribers.}$$

f.) If each cell covers five square kilometers, then how many subscribers could be supported in an urban market that is $50\text{ km} \times 50\text{ km}$ for the case of 120° sectored antennas

$\rightarrow$ Using approximately 765 users/cell :

$$500 \times 765 = 382,500 \text{ subscribers.}$$

3) →

Given,

$$\text{Total spectrum} = 20 \text{ MHz}$$

$$\text{Simplex channel} = 25 \text{ kHz}$$

$$\text{Reuse factor} = 4$$

a)

→ A duplex channel requires two simplex channels :

$$2 \times 25 \text{ kHz} = 50 \text{ kHz}$$

$$\therefore \frac{20 \text{ MHz}}{50 \text{ kHz}}$$

$$= \frac{20,000}{50}$$

$$= 400 \text{ duplex channel}$$

b)

→ channels per cell :

$$\frac{400}{4}$$

$$= 100 \text{ channels per cell.}$$

4,

→

Given,

Number of cells = 84

Total channels = 300

Traffic / user = 0.04 Erlang

Blocking Probability = 1%

$N = 4, 7, 12$

a

→ For $N = 4$,

$$C = \frac{300}{4}$$

= 75 channels.

For 75 channels at 1% blocking,

$A \approx 60.73$ Erlangs.

For $N = 7$

$$C = \frac{300}{7}$$

= 42 channels

$A \approx 30.77$ Erlangs

For $N = 12$

$$C = \frac{300}{12}$$

= 25 channels

$A \approx 16.12$ Erlangs.

b)

→

$$U = \frac{A}{0.04}$$

For $N = 4$

$$U = \frac{60.73}{0.04} \approx 1518$$

For 84 cells,

$$1518 \times 84 \approx 127,528 \text{ users}$$

For $N = 7$

$$U = \frac{30.77}{0.04} \approx 769$$

$$769 \times 84 \approx 64,620 \text{ users}$$

For $N = 12$

$$U = \frac{16.12}{0.04} \approx 403$$

$$403 \times 84 \approx 33,862 \text{ users}$$

5)

→ a)

$$R = \frac{270.833}{8}$$

$$\therefore R = 33.85 \text{ kbps/user}$$

b)

$$\begin{aligned}\text{Useful rate} &= 33.854 - 10.1 \\ &= 23.754 \text{ kbps}\end{aligned}$$

Efficiency,

$$\eta = \frac{23.754}{33.854} \times 100$$

$$\therefore \eta = 70.16\%$$

6)

a)

Since, channels are numbered 1, 2, 3, ..., 20

$\therefore$ There are 20 channels.

b)

Channels 1 to 12 are assigned to cell 1 to 12 ~~are assi~~ respectively using FCA Strategy.

Channels 13-20 can be kept as additional/overflow channel and assigned according to traffic requirements.

7)

→ Features of DECT are ÷

- i) Digital Communication → provides good voice quality.
- ii) Cordless Operation → allows mobility within the coverage area.
- iii) low-power transmission → suitable for indoor applications.
- iv) Multiple users → TDMA/TDD technology allows multiple users to share radio resources.
- v) Security → Supports authentication and encryption mechanisms.
- vi) Roaming → Supports mobility between compatible base station.
- vii) Good voice quality & data transmission reduces interference and noise.

DECT commonly operates in the 1880-1900 MHz frequency band in Europe

8)

→ Given,

$$\text{No. of cells} = 32$$

$$\text{Radius} = 1.6 \text{ km}$$

$$\text{Total traffic channels} = 336$$

$$\text{Reuse factor} = 7$$

a)

→ We know,

$$A = \frac{3\sqrt{3}}{2} R^2 \quad (\text{for hexagon})$$

$$= \frac{3\sqrt{3}}{2} (1.6)^2$$

$$A \approx 6.651 \text{ km}^2$$

For 32 cells,

$$A_{\text{total}} = 6.651 \times 32$$

$$A_{\text{total}} \approx 212.84 \text{ km}^2$$

b)

$$\rightarrow \text{channels rate} = \frac{336}{7}$$

$$= 48 \text{ channels/cell}$$

c)

→ Each channel can support one concurrent call.

$$48 \times 32 = 1536 \text{ concurrent calls.}$$

g) ISDN has been more widely accepted as a network
→ ISDN solution, there are various types of ISDN; clarify it, include speed, uses.

→ ISDN (Integrated Services Digital Network) is a digital telecommunication network that allows voice, data, video and fax to be transmitted through telephone lines. It provides faster and more reliable communication than traditional analog telephone systems.

The main types of ISDN are BRI, PRI and B-ISDN.

- BRI (Basic Rate Interface) → Provides 2 B channels of 64 kbps each and 1 D channel of 16 kbps, with a total speed of 144 kbps. It is mainly used in homes and small offices.

- PRI (Primary Rate Interface) → Provides 1.544 Mbps using T1 (23B + 1D) or 2.048 Mbps using E1 (30B + 1D). It is managed mainly used by large organizations and businesses.

- B-ISDN (Broadband ISDN) $\rightarrow$ Provides very high-speed communication, generally 155 Mbps or higher, and is designed for multimedia, video and other high bandwidth services.

* Uses of ISDN :

- Digital telephone and voice communication.
- Internet and data transmission.
- Video Conferencing
- Fax communication
- Business telephone and PBx systems
- Transmission of multimedia and video services.

10)

$\rightarrow$ Given,

forward-link bandwidth = 20 MHz

Radio channels = 20 kHz

Speech channel per radio channels = 6

$$\text{No. of radio channels} = \frac{20 \text{ MHz}}{20 \text{ kHz}}$$

$$= \frac{20,000}{20}$$

$$= 1000 \text{ radio channel}$$

Each radio supports 6 speech channels : 1000×6

= 600 simultaneous users.

11) Describe the key features and functions of signaling system No. 7 (SS7) in telecommunication networks. Explain how SS7 facilitates call setup, routing, and termination processes. Discuss the advantages of using SS7 in comparison to in-band signaling techniques.

→ Signaling system No. 7 (SS7) is a set of telecommunication signaling protocols used to control and manage telephone calls and other services in a telecommunication network. It uses a separate signaling network, called an out-of-band signaling network, to exchange control information rather than sending signaling tones through the voice channel.

* key features and functions of SS7 :

- Out-of-band signaling → Signaling information is transmitted separately from the voice/data path.
- Call setup and control → It establishes, controls and terminates telephone calls.
- Call routing → SS7 helps determine the appropriate route and destination for a

call.

- Call termination → It sends signals to release network resources when a call ends.
- Number translation → It translates telephone numbers and helps identify the correct destination.
- Database access → It allows network services to access databases for services such as toll-free numbers, caller identification, and mobile services.
- Error detecting and reliability → SS7 includes mechanisms for detecting errors and maintaining reliable signaling.
- Supports advanced services → It supports services such as call forwarding, roaming, SMS, caller ID, and prepaid services.

* SS7 facilitates a call by :-

- Call setup → When a user makes a call, SS7 signaling messages containing the calling and called numbers to the network.
- Routing → The network analyzes the signaling information and selects the appropriate route to connect the call to destination.

- Call Establishment → Once the destination is reached and the called party answers, SS7 confirms the connection and allows the voice path to carry the conversation.
- Call termination: When either party ends the call, SS7 sends a release message, causing the network to disconnect the call and free the resources.
- * Advantages of SS7 over In-Band Signaling,
 - Provides faster and more efficient signaling.
 - Signaling is independent of the voice channel, so it does not interfere with the conversation.
 - Provides better security and reliability than traditional in-band signaling.
 - Allows signaling to occur before the voice connection is established, making call setup faster.
 - Supports advanced and intelligent network services.
 - Provides better error detection and network management.

12)

→ Given,

new calls per hour = 1800

Average holding time = 3 minutes

Traffic density :

$$A = \lambda H$$

Converting holding time into hours,

$$H = \frac{3}{60}$$

$$= 0.05 \text{ hour}$$

$$\therefore A = 1800 \times 0.05$$
$$= 90 \text{ Erlangs}$$

13)

→ Given,

Total bandwidth = 33 MHz

Simplex channel = 25 kHz

$$N = 4, 7, 12$$

one duplex channel requires :

Date _____
Page No. _____

$$2 \times 25$$
$$= 50 \text{ KHz}$$

$$\therefore \text{Total duplex channel} = \frac{33 \text{ MHz}}{50 \text{ KHz}}$$

$$= \frac{33000}{50}$$

$$= 660 \text{ duplex channel}$$

For $N=4$,

$$\frac{660}{4}$$
$$= 165 \text{ channels/cell}$$

For $N=7$,

$$\frac{660}{7}$$
$$= 94 \text{ channels/cell}$$

For $N=12$,

$$\frac{660}{12} = 55 \text{ channels/cell}$$

14)

→ Given,

$$R = 48.6 \text{ kbps}$$

$$\text{No. of users} = 3$$

Raw data per user ;

$$R_u = \frac{48.6}{3}$$

$$= 16.2 \text{ kbps/user.}$$